

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
dm=pd.read_csv(r"C:\Users\Maharshi Goswami\Downloads\dataminig\marketing_campaign.csv",sep='\t')
```

```
dm.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
0	5524	1957	Graduation	Single	58138.0	0
1	2174	1954	Graduation	Single	46344.0	1
2	4141	1965	Graduation	Together	71613.0	0
3	6182	1984	Graduation	Together	26646.0	1
4	5324	1981	PhD	Married	58293.0	1

	Dt_Customer	Recency	MntWines	...	NumWebVisitsMonth	AcceptedCmp3
0	04-09-2012	58	635	...	7	0
1	08-03-2014	38	11	...	5	0
2	21-08-2013	26	426	...	4	0
3	10-02-2014	26	11	...	6	0
4	19-01-2014	94	173	...	5	0

	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	\
0	0	0	0	0	0	
1	0	0	0	0	0	
2	0	0	0	0	0	
3	0	0	0	0	0	
4	0	0	0	0	0	

	Z_CostContact	Z_Revenue	Response
0	3	11	1
1	3	11	0
2	3	11	0
3	3	11	0
4	3	11	0

```
[5 rows x 29 columns]
```

Cleaning

```
dm.isnull().sum()/len(dm)*100
```

```
ID          0.000000
Year_Birth  0.000000
Education   0.000000
Marital_Status 0.000000
Income      1.071429
Kidhome     0.000000
Teenhome    0.000000
Dt_Customer 0.000000
Recency     0.000000
MntWines    0.000000
MntFruits   0.000000
MntMeatProducts 0.000000
MntFishProducts 0.000000
MntSweetProducts 0.000000
MntGoldProds 0.000000
NumDealsPurchases 0.000000
NumWebPurchases 0.000000
NumCatalogPurchases 0.000000
NumStorePurchases 0.000000
NumWebVisitsMonth 0.000000
AcceptedCmp3 0.000000
AcceptedCmp4 0.000000
AcceptedCmp5 0.000000
AcceptedCmp1 0.000000
AcceptedCmp2 0.000000
Complain    0.000000
Z_CostContact 0.000000
Z_Revenue   0.000000
Response    0.000000
dtype: float64
```

```
# Income missing values
```

```
dm['Income'] = dm['Income'].fillna(dm['Income'].median())
```

Total Spending features

```
# Missing values
```

```
print(dm.isnull().sum())
```

```
# Duplicates
```

```
dm.drop_duplicates(inplace=True)
```

```
# Total spending
```

```
dm['Total_Spending'] = (
```

```

    dm['MntWines'] +
    dm['MntFruits'] +
    dm['MntMeatProducts'] +
    dm['MntFishProducts'] +
    dm['MntSweetProducts'] +
    dm['MntGoldProds']
)
ID                                0
Year_Birth                       0
Education                       0
Marital_Status                  0
Income                          0
Kidhome                         0
Teenhome                        0
Dt_Customer                     0
Recency                         0
MntWines                        0
MntFruits                       0
MntMeatProducts                 0
MntFishProducts                 0
MntSweetProducts                0
MntGoldProds                    0
NumDealsPurchases               0
NumWebPurchases                 0
NumCatalogPurchases            0
NumStorePurchases               0
NumWebVisitsMonth               0
AcceptedCmp3                    0
AcceptedCmp4                    0
AcceptedCmp5                    0
AcceptedCmp1                    0
AcceptedCmp2                    0
Complain                        0
Z_CostContact                   0
Z_Revenue                       0
Response                        0
dtype: int64

```

Feature Selection

```

features = dm[[
    'Income',
    'Recency',
    'Total_Spending',
    'NumWebPurchases',
    'NumCatalogPurchases',
    'NumStorePurchases'
]]

```

Feature Scaling

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X = scaler.fit_transform(features)
```

KMEANS APPLY

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
dm['Cluster'] = kmeans.fit_predict(X)
print(dm[['Cluster']].head())
```

	Cluster
0	3
1	2
2	1
3	2
4	0

```
print(dm['Cluster'].value_counts())
```

Cluster	
1	583
2	576
3	542
0	539

Name: count, dtype: int64

```
cluster_summary = dm.groupby('Cluster')[[
    'Income',
    'Total_Spending',
    'Recency'
]].mean()
```

```
print(cluster_summary)
```

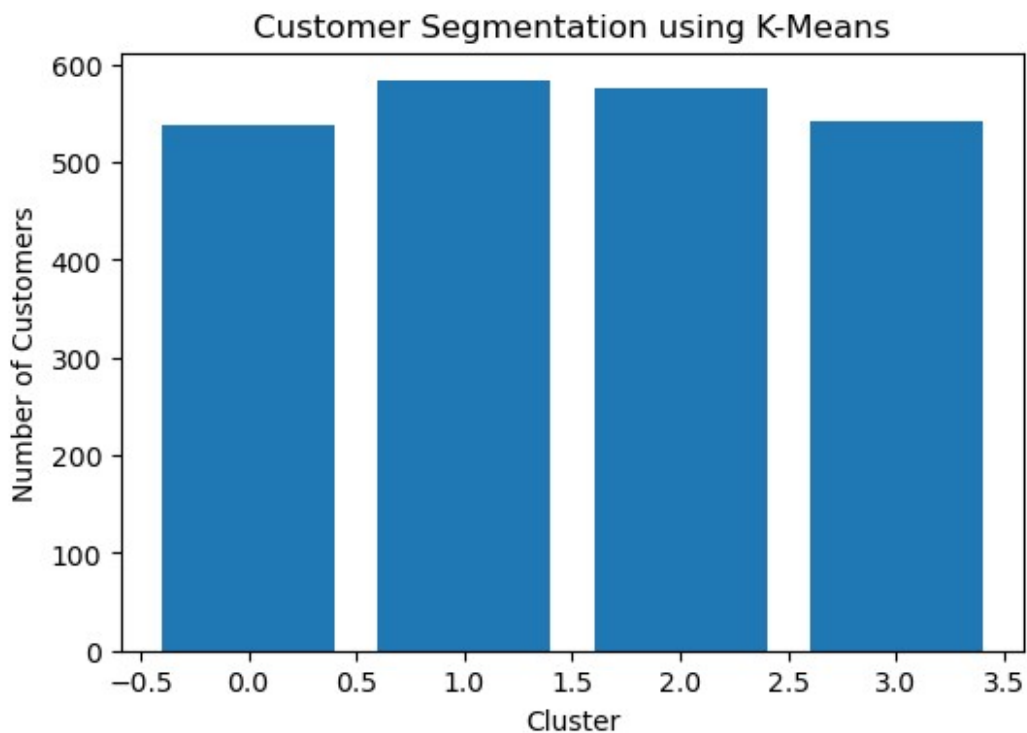
	Income	Total_Spending	Recency
Cluster			
0	37491.525974	135.428571	76.608534
1	60021.775300	764.591767	42.481990
2	34687.046007	97.753472	25.581597
3	77182.101476	1442.673432	53.894834

EDA

```
import matplotlib.pyplot as plt

cluster_counts = dm['Cluster'].value_counts().sort_index()
```

```
plt.figure(figsize=(6,4))
plt.bar(cluster_counts.index, cluster_counts.values)
plt.xlabel('Cluster')
plt.ylabel('Number of Customers')
plt.title('Customer Segmentation using K-Means')
plt.show()
```

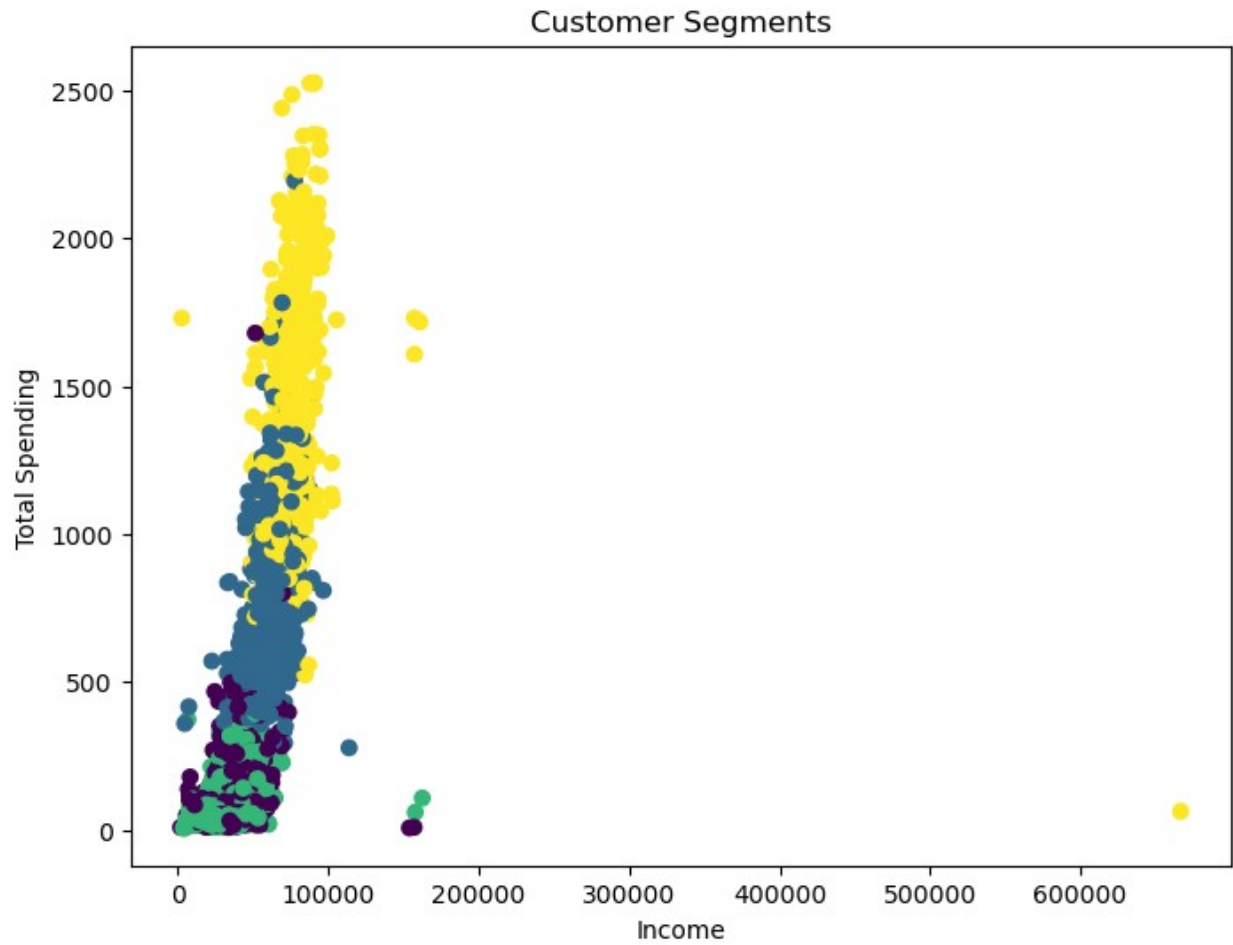


K-Means clustering successfully divided customers into four meaningful groups, with each cluster containing a comparable number of customers. This indicates that the segmentation is balanced and suitable for targeted marketing strategies.

```
plt.figure(figsize=(8,6))

plt.scatter(
    dm['Income'],
    dm['Total_Spending'],
    c=dm['Cluster']
)

plt.xlabel('Income')
plt.ylabel('Total Spending')
plt.title('Customer Segments')
plt.show()
```



Customers with higher income tend to spend more on products, indicating a positive relationship between income and purchasing behavior.